Evaluating Continuity-Based Physics Regularization in Fourier Neural Operators for Geometric Extrapolation of Aerodynamic Flow Fields

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1. Introduction

Neural operators can predict aerodynamic flow fields significantly faster than conventional Computational Fluid Dynamics (CFD) solvers. However, low overall field error does not necessarily guarantee that predictions are physically consistent. This project investigates whether adding a **continuity-based physics regularization term** to a **Fourier Neural Operator (FNO)** can reduce local physical inconsistencies when predicting flow over **unseen blade geometries**.

1.1 The Engineering Problem

Purely data-driven surrogate models (like a standard FNO) minimize global structural error (Relative L_2 against CFD ground-truth) but remain entirely unaware of the underlying physics. As a result, they often produce velocity fields that violate local continuity (conservation of mass), especially when extrapolating.

By explicitly penalizing non-physical divergence during training, a Physics-Informed FNO (PI-FNO) forces the network to respect fluid dynamics equations. The core question is whether this physics regularization can improve physical consistency without causing an unacceptable loss of predictive accuracy.

2. Experimental Protocol

To ensure the integrity of the comparison and avoid overfitting the physics regularization term to the test data, the dataset was strictly partitioned.

2.1 Dataset Partitioning

- **Training Set:** 100 geometries. Used to fit the network weights via Adam optimization.
- **Validation Set:** 25 geometries. Used exclusively to evaluate different candidate values for the physics loss weighting parameter (λ).
- **Test Set:** 50 geometries. Held out completely until the final models were locked.

2.2 Model Selection

The baseline FNO ($\lambda = 0$) and several PI-FNO candidates were trained. The validation set was used to select the optimal physics weight that maximized continuity improvement without causing an unacceptable degradation in global L_2 accuracy. This process identified **λ = 1.0** as the optimal weight for this dataset.

2.3 Held-Out Evaluation

The final baseline FNO and the selected PI-FNO ($\lambda = 1.0$) were evaluated exactly once on the 50 held-out test geometries. The resulting metrics form the definitive performance claims of this project.

2.4 Low-Data Analysis

To understand how physics constraints impact performance when data is scarce, an additional experiment was run using only 25 training geometries. This experiment was repeated across three random seeds to ensure the resulting accuracy penalty (≈ 0.00175) was stable and not a statistical anomaly.

3. Methodology

This project implements a controlled comparison between a baseline Fourier Neural Operator (FNO) and a Physics-Informed Fourier Neural Operator (PI-FNO) for aerodynamic flow-field prediction.

3.1 Network Architecture

Both models use the identical underlying 3D FNO architecture, initialized with:

- 8 Fourier modes across all three spatial dimensions.
- A channel width of 16.
- A standard MSE loss formulation for the data-driven component.

3.2 Physics Regularization

The baseline FNO is trained purely on the data-driven MSE loss, optimizing for global structural accuracy (Relative L_2).

For the PI-FNO, I introduced a custom PDE residual loss based on the continuity equation (conservation of mass). The continuity residual is defined as the discrete spatial divergence of the density-velocity flux:
